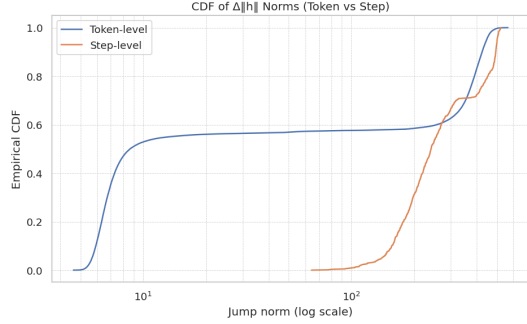
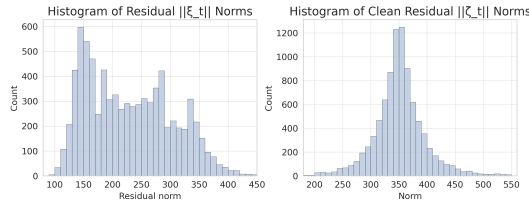




(a)



(b)

Figure 1. (a) CDF comparison of token and sentence jump norms, illustrating that sentence-level increments capture more substantial semantic shifts. (b) Histograms of residual norms from a global linear fit, showing raw residuals $\|\xi_t\|$ (left) and residuals projected onto a low-rank PCA space $\|\zeta_t\|$ (right). Both reveal significant multimodality, motivating regime switching to capture distinct reasoning phases or potential misalignments.

these regimes. Such switches may represent transitions between different cognitive states, some of which could be misaligned or lead to errors.

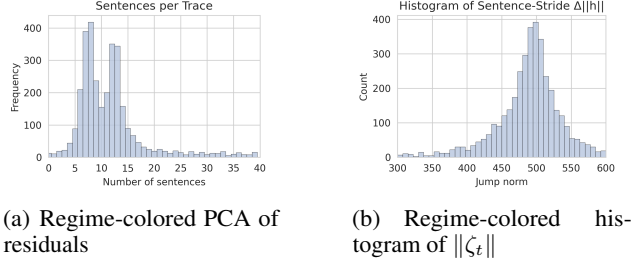
4.1. Linear Drift within Regimes

While a single global linear model (Eq. 5) captures about half the variance, the residual analysis (Figure 1(b)) indicates that a more nuanced approach is needed. We project the residuals ξ_t onto the principal subspace V_k (from Assumption 2.1, where $k = 40$ offers a balance between explained variance and computational cost) to get $\zeta_t = V_k^\top \xi_t$. The clustered nature of these projected residuals ζ_t suggests that the reasoning process transitions between several distinct dynamical modes or ‘regimes’.

4.2. Identifying Latent Reasoning Regimes

To formalize these distinct modes, we fit a K -component Gaussian Mixture Model (GMM) to the projected residuals ζ_t , following classical regime-switching frameworks (Hamilton, 1989):

$$p(\zeta_t) = \sum_{i=1}^K \pi_i \mathcal{N}(\zeta_t \mid \mu_i, \Sigma_i). \quad (7)$$



(a) Regime-colored PCA of residuals

(b) Regime-colored histogram of $\|\zeta_t\|$

Figure 2. Latent regimes ($K = 4$) uncovered by GMM fitting on low-rank residuals ζ_t . (a) Residuals projected onto their first two principal components, colored by GMM assignment, showing distinct clusters. (b) Histogram of residual norms $\|\zeta_t\|$, colored by GMM regime assignment, further illustrating regime separation. These regimes may capture different reasoning qualities, including potential misalignments.

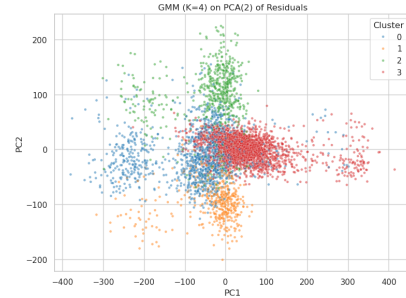


Figure 3. GMM clustering ($K = 4$) of low-rank residuals ζ_t , visualized in the space of the first two principal components of ζ_t . The distinct cluster centers provide justification for the regime decomposition, potentially corresponding to different reasoning states or failure modes.

Information criteria (BIC/AIC) suggest $K = 4$ as an appropriate number of regimes for our data. While the true underlying multimodality is complex across many dimensions (see Figure 6, Appendix A), a four-regime model provides a parsimonious yet effective way to capture key dynamic behaviors, including those that might represent misalignments or slips into undesired reasoning patterns, while maintaining computational tractability. We interpret these $K = 4$ modes as distinct reasoning phases, such as systematic decomposition, answer synthesis, exploratory variance, or even failure loops, each characterized by specific drift perturbations and noise profiles. Figure 2 and Figure 3 visualize these uncovered regimes in the low-rank residual space.

4.3. The Switching Linear Dynamical System (SLDS) Model

We integrate these observations into a discrete-time Switching Linear Dynamical System (SLDS). Let $Z_t \in$

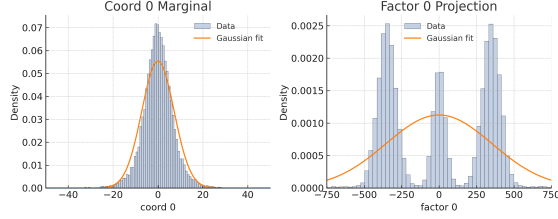


Figure 4. Failure of single-mode noise models for the full residuals ξ_t (before projection). This plot shows mismatches between the empirical distribution of residual norms and fits from both Gaussian and Laplace distributions, highlighting the inadequacy of a single noise process and further motivating the regime-switching approach to capture diverse reasoning states, including potential misalignments.

$\{1, \dots, K\}$ be the latent regime at step t . The state h_t evolves according to:

$$\begin{aligned} Z_t &\sim \text{Categorical}(\pi), \quad P(Z_{t+1} = j \mid Z_t = i) = T_{ij}, \\ h_{t+1} &= h_t + V_k(M_{Z_t}(V_k^\top h_t) + b_{Z_t}) + \varepsilon_t, \\ \varepsilon_t &\sim \mathcal{N}(0, \Sigma_{Z_t}). \end{aligned} \quad (8)$$

Here, $M_i \in \mathbb{R}^{k \times k}$ and $b_i \in \mathbb{R}^k$ are the regime-specific linear transformation matrix and offset vector for the drift within the k -dimensional semantic subspace defined by V_k . Σ_i is the regime-dependent covariance for the noise ε_t . The initial regime probabilities are π , and T is the transition matrix encoding regime persistence and switching probabilities. This SLDS framework combines continuous drift within regimes, structured noise, and discrete changes between regimes, which can model shifts between correct reasoning and misaligned states.

The multimodal structure of the full residuals ξ_t (before projection, see Figure 4) invalidates a single-mode SDE. This motivates our regime-switching formulation. The SLDS in Eq. 8 serves as a discrete-time surrogate for an underlying continuous-time switching SDE (Eq. 4):

$$dh(t) = \mu_{Z(t)}(h(t)) dt + B_{Z(t)}(h(t)) dW(t), \quad (9)$$

where each regime i has its own drift $\mu_i(h) = V_k(M_i(V_k^\top h) + b_i)$ (approximating the continuous drift within the chosen manifold for tractability) and diffusion B_i (related to Σ_i). The transition matrix T in the SLDS is related to the rate matrix of the latent Markov process $Z(t)$ in the continuous formulation.

5. Experiments & Validation

We empirically validate the proposed SLDS framework (Eq. 8). Our primary goal is to demonstrate that this model, operating on a practically chosen low-rank manifold, can effectively learn and represent the general dynamics of

sentence-level semantic evolution, including transitions that might signify a slip into misaligned reasoning. The SLDS parameters $(\{M_i, b_i, \Sigma_i\}_{i=1}^K, T, \pi)$ are estimated from our corpus of $\sim 40,000$ sentence-to-sentence hidden state transitions using an Expectation-Maximization (EM) algorithm (Appendix B). It is crucial to note that the SLDS is trained to model the *process* by which language models arrive at answers—and potentially how they deviate into failure modes—not to predict the final answers of the tasks themselves. Based on empirical findings (Section 4), we use $K = 4$ regimes and a projection rank $k = 40$ (chosen for its utility in making the SDE-like modeling feasible).

The efficacy of the fitted SLDS is first assessed by its one-step-ahead predictive performance. Given an observed hidden state h_t and the inferred posterior regime probabilities $\gamma_{t,j} = \mathbb{P}(Z_t = j \mid h_0, \dots, h_t)$ (obtained via forward-backward inference (Rabiner, 1989)), the model’s predicted mean state $\hat{h}_{t+1}$ is computed as:

$$\hat{h}_{t+1} = h_t + V_k \left(\sum_{j=1}^K \gamma_{t,j} (M_j(V_k^\top h_t) + b_j) \right). \quad (10)$$

On held-out trajectories, the SLDS yields a predictive $R^2 \approx 0.68$. This significantly surpasses the $R^2 \approx 0.51$ achieved by the single-regime global linear model (Eq. 5), confirming the value of incorporating regime-switching dynamics. Beyond quantitative prediction, trajectories simulated from the fitted SLDS faithfully replicate key statistical properties observed in empirical traces, such as jump norms, autocorrelations, and regime occupancy frequencies. This dual capability—accurate description and realistic synthesis of reasoning trajectories—substantiates the SLDS as a robust model. Furthermore, the inferred regime posterior probabilities $\gamma_{t,j}$ provide valuable interpretability, allowing for the association of observable textual behaviors (e.g., systematic decomposition, stable reasoning, or error correction loops and potential misaligned states) with specific latent dynamical modes. These initial findings strongly support the proposed framework as both a descriptive and generative model of reasoning dynamics, offering a path to predict and understand LLM failure modes.

5.1. Generalization and Transferability of SLDS Dynamics

A critical test of the SLDS framework is its ability to capture generalizable features of reasoning dynamics, including those indicative of robust reasoning versus slips into misalignment, beyond the specific training conditions. We investigated this by training an SLDS on hidden state trajectories from a *source* (a particular LLM performing a specific task or set of tasks) and then evaluating its capacity to describe trajectories from a *target* (which could be a different LLM and/or task). Transfer performance was

quantified using two metrics: the one-step-ahead prediction R^2 for the projected hidden states (Eq. 10) and the Negative Log-Likelihood (NLL) of the target trajectories under the source-trained SLDS. Lower NLL and higher R^2 values signify superior generalization.

Table 1 presents illustrative results from these transfer experiments. For instance, an SLDS is first trained on trajectories generated by a ‘Train Model’ (e.g., Llama-2-70B) performing a designated ‘Source Task’ (e.g., GSM-8K). This single trained SLDS is then evaluated on trajectories from various ‘Test Model’ / ‘Test Task’ combinations. The results indi-

Table 1. SLDS transferability across models and tasks. Each SLDS is trained on trajectories from the specified ‘Train Model’ on its ‘Source Task’ (GSM-8K for Llama-2-70B, StrategyQA for Mistral-7B). Performance (R^2 for next hidden state prediction, NLL of test trajectories) is evaluated on various ‘Test Model’ / ‘Test Task’ combinations, demonstrating patterns of generalization in capturing underlying reasoning dynamics.

TRAIN MODEL (SOURCE TASK)	TEST MODEL	TEST TASK	R^2	NLL
LLAMA-2-70B (ON GSM-8K)	LLAMA-2-70B	GSM-8K	0.73	80
	LLAMA-2-70B	STRATEGYQA	0.65	115
	MISTRAL-7B	GSM-8K	0.48	240
	MISTRAL-7B	STRATEGYQA	0.37	310
MISTRAL-7B (ON STRATQA)	MISTRAL-7B	STRATEGYQA	0.71	88
	MISTRAL-7B	GSM-8K	0.63	135
	LLAMA-2-70B	STRATEGYQA	0.42	270
	GEMMA-7B-IT	BOOLQ	0.35	380
	PHI-3-MED	TRUTHFULQA	0.30	420

cate that while the SLDS performs optimally when training and testing conditions align perfectly (e.g., Llama-2-70B on GSM-8K transferred to itself), it retains considerable descriptive power when transferred. Generalization is notably more successful when the underlying LLM architecture is preserved, even across different reasoning tasks (e.g., Llama-2-70B trained on GSM-8K and tested on StrategyQA shows only a modest drop in R^2 from 0.73 to 0.65). Conversely, transferring the learned dynamics across different LLM families (e.g., Llama-2-70B to Mistral-7B) proves more challenging, as reflected in lower R^2 values and higher NLLs. However, even in these challenging cross-family transfers, the SLDS often outperforms naive baselines like a simple linear dynamical system without regime switching (detailed comparisons not shown). These findings suggest that while some learned dynamical features are model-specific, the SLDS framework, by approximating the reasoning process as a physicist might model a complex system, is capable of capturing common, fundamental underlying structures in reasoning trajectories. Extended transferability results are provided in Appendix D.

5.2. Ablation Study

To elucidate the contribution of each core component within our SLDS framework, we conducted an ablation study. The full model (Eq. 8 with $K = 4$ regimes and $k = 40$ projection rank, selected for practical modeling of the SDE) was compared against three simplified variants:

- **No Regime (NR):** A single-regime model ($K = 1$), still projected to the $k = 40$ dimensional subspace. This tests the necessity of regime switching for capturing diverse reasoning states, including misalignments.
- **No Projection (NP):** A $K = 4$ regime switching model operating directly in the full D -dimensional embedding space (i.e., without the V_k projection). This tests the utility of the low-rank manifold assumption for tractable and effective modeling, given the impracticality of handling a full-dimension SDE.
- **No State-Dependent Drift (NSD):** A $K = 4$ regime model where the drift within each regime is merely a constant offset $V_k b_{Z_t}$, and the linear transformation M_{Z_t} is zero for all regimes. This tests the importance of the current state h_t influencing its own future evolution within a regime.

Table 2 summarizes the performance of these models on a held-out test set. Each ablation led to a notable reduction

Table 2. Ablation study results comparing the full SLDS against simplified variants: NR (single-regime projected model), NP (full-dimensional switching without projection), NSD (regime-switched offsets, no state-dependent linear drift). Performance is measured by R^2 and NLL. The results underscore the importance of each component for modeling reasoning dynamics and identifying potential failure modes.

MODEL	R^2	NLL
FULL SLDS ($K = 4, k = 40$)	0.74	78
NO REGIME (NR, $K = 1, k = 40$)	0.58	155
NO PROJECTION (NP, $K = 4$)	0.60	210
NO STATE-DEP. DRIFT (NSD)	0.35	290
<i>Global Linear (ref.)</i>	<i>0.51</i>	<i>180</i>

in performance, robustly demonstrating that all three key elements of our proposed model—regime-switching, low-rank projections (for practical SDE approximation), and state-dependent drift—are jointly essential for accurately capturing the nuanced dynamics of transformer reasoning. The NR model, lacking regime switching, performs substantially worse ($R^2 = 0.58$) than the full SLDS ($R^2 = 0.74$), highlighting the critical role of modeling distinct reasoning phases, including potential slips into misaligned states. Removing the low-rank projection (NP model) also significantly impairs effectiveness ($R^2 = 0.60$), suggesting that attempting to learn high-dimensional drift dynamics